Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 State the type of control and describe why it is used.

- AND control
- Both valves must be actuated, e.g. two-handed safety so the operator's hands are clear

3 Quick check

- a) AND control
- b) 113 mm²
- c) No sparks, so safe near flammable materials
- d) 5 ports, 2 states

4 Key terms

- 1 – A
- 2 – F
- 3 – B
- 4 – C
- 5 – E
- 6 – D

5 Calculate the outstroke force.

- $F = p \times A = 0.5 \times 800$
- $F = 400$ N

6 Describe how the circuit can be modified.

- Add a unidirectional flow control valve (restrictor) to the exhaust
- Restricting the air leaving the cylinder slows the outstroke smoothly

7 Describe what happens when valve A is actuated.

- Valve A sends a pilot signal and the 5/2 valve changes state
- Main air goes to the rear of the cylinder and the piston outstrokes